



Fig. 7: The device after setting and fixing all components

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File Edit Format Run Options Window Help
from Adafruit_AMG88xx import Adafruit_AMG88xx
import pygame
import os
import math
import time

import numpy as np
from scipy.interpolate import griddata

from colour import Color

#low range of the sensor (this will be blue on the screen)
MINTEMP = 25

#high range of the sensor (this will be red on the screen)
MAXTEMP = 36

#how many color values we can have
COLORDEPTH = 1024

os.putenv('SDL_FBDEV', '/dev/fb1')
pygame.init()

#initialize the sensor
sensor = Adafruit_AMG88xx()

points = [(math.floor(ix / 8), (ix % 8)) for ix in range(0, 64)]
grid_x, grid_y = np.mgrid[0:7:32j, 0:7:32j]

#sensor is an 8x8 grid so lets do a square
height = 600
width = 600

#the list of colors we can choose from
blue = Color("indigo")
colors = list(blue.range_to(Color("red"), COLORDEPTH))
```

Fig. 8: Code setting the display size and sensor

camera has a thermal sensor inside many small IR thermal sensors. Each thermal sensor acts as a pixel of the thermal camera and it is in the form of a matrix. So, if a good-quality, high-resolution thermal camera sensor is used, it has very small thermal sensors in the form of pixels.

Here, we are using the AMG8833 thermal sensor, so the pixel size is a matrix of 8×8 . To get the real-time video, we map those pixels, that

Testing

When the device is ready after preparing the above code, connect it to the charging port/USB of your iPhone or Android phone. Install the VNC viewer on the phone or PC on which

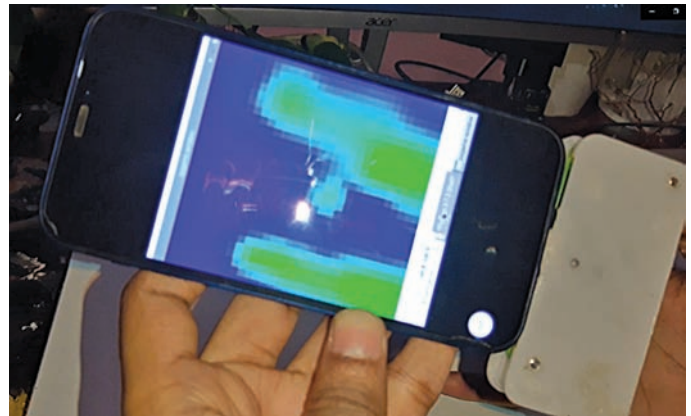


Fig. 9: Testing the thermal camera on phone

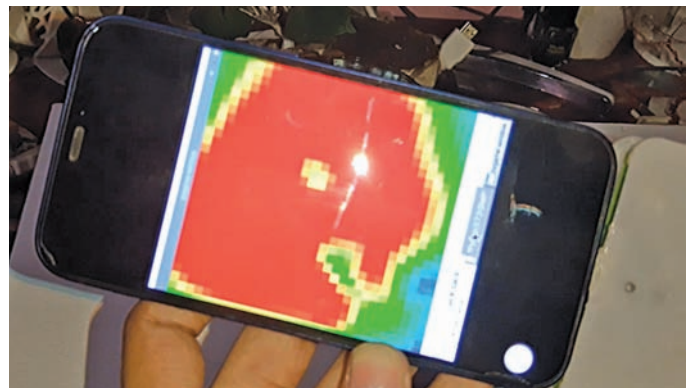


Fig. 10: Output displayed on phone screen

is, the small-sized IR thermal sensor data, with colours like blue, yellow, and red, where red depicts the hottest and blue the coldest. After colour mapping, the thermal image is formed that we see in any thermal camera video.

Fig. 8 shows the code setting the display size and sensor.

you want to receive the thermal camera video.

If iPhone is being converted to a thermal camera by attaching it to this device, install the VNC viewer, available on the app store, on the iPhone. We need to insert the IP address and password of our Raspberry Pi and let the Pi connect to either a phone hotspot or to the same Wi-Fi network that our phone is connected to.

Now run the code that was prepared earlier to get the real-time thermal video. Move your hand near the thermal sensor device, or bring a hot object near the sensor device, and you can see the thermal video of the object.

You have just designed the thermal screening attachment accessory for the phone. Fig. 9 and Fig. 10 show testing of the project. **EFY**

EFY Note

The source code of this project is available for free download at source.efymag.com

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